Assignment 02

Department of Applied Mathematics, University of Dhaka

Second Year BS (Honors), 2025-26

Course Title: Math Lab II, Course Code: AMTH-250

Name: **Roll:** **Group:**

Q1. Show that $f(x) = x^3 - 4x - 9 = 0$ has a root in $[2,\ 3]$. Use the **Bisection** method to determine an approximation to the root that is accurate to at least within 10^{-4}. Show your answer in a table with heading as follows:

$$Iteration\ NO. \quad a \quad b \quad P_n \quad \frac{|P_n - P_{n-1}|}{|P_n|}$$

Q2. Use the **Secant Method** to find an approximate root of $e^{-x} - x = 0$ in the interval $[0, 1]$, correct to **six decimal places**. Show your results in a tabular form using suitable table headings.

Q3. Use the **False Position (Regula Falsi) Method** to find a solution of $\ln x + x^2 - 3 = 0$ in the interval $[1, 2]$, Continue the iterations until the answer is accurate within 10^{-5}. Display your results in a suitable table as:

$$Iteration\ NO. \quad a \quad b \quad P_n \quad |P_n - P_{n-1}|$$

Q4. Use the **Fixed Point Iteration** method to determine an approximation to the root of the equation $x^3 + x - 1 = 0$ on $[0,1]$ that is accuarate to within 10^{-4}. Take $g(x) = \frac{1}{1+x^2}$. Show your answer in a table with headings as follows:

$$Iteration\ NO. \quad P_{n-1} \quad P_n \quad \frac{|P_n - P_{n-1}|}{|P_n|}$$

Q5. Find the real root of the equation $e^x - 3x = 0$ correct to five decimal places, by **Newton-Raphson** method considering the following initial guesses:

i) $x_0 = 0.0$
ii) $x_0 = -1.0$
iii) $x_0 = 2.0$
iv) $x_0 = -2.50$

Hence, show your findings in three separate tables each containing headings as follows:

$$Iteration\ NO. \quad P_{n-1} \quad P_n \quad \frac{|P_n - P_{n-1}|}{|P_n|}$$